

# Eric J. Leonardis, PhD

San Diego, CA. 92110

Email: leonardiseric@gmail.com

Phone: 516-510-2096

## EDUCATION AND TRAINING

**Salk Institute for Biological Studies** – August 2022 to Present

**Postdoctoral Fellow** – NIH T32 Trainee

**Research Interests:** Deep Reinforcement Learning, 3D Pose Tracking

**Project** - “Musculoskeletal Modeling and Imitation Learning for Dexterous Reaching with the Mouse Forelimb.”

**Grants Awarded:**

**“Interpretable NeuroAI for Motor Control: Massively Parallel Imitation Learning of Musculoskeletal Dynamics”** – \$100,000

Foresight Institute Grant – October 2025 to Present

**Grants Contributed To:**

**“Understanding Sensorimotor Control Through Realistic Neuro-Biomechanical Simulation”** – \$2,478,214

NIH BRAIN U01 Grant – Aug 2023 to October 2025

**“Collaborative Research: Perception, Behavior and Learning in the Museum”** – \$530,819

NSF Grant – Aug 2022 to Aug 2023

**University of California, San Diego** – September 2014 to June 2022

**PhD and MS in Cognitive Science**

**Dissertation** - “Interactive Neurorobotics: Brain and Body Coupling During Interactive Multi-Agent Scenarios”

**Research Interests:** Systems Neuroscience, Robotics, Behavior

National Science Foundation Temporal Dynamics of Learning Center Trainee

**GPA:** 3.98

**Grants Awarded:**

**“Discovering patterns in human-robot interaction: New tools for complex adaptive social systems”** \$300,000

US-AU Air Force Office of Scientific Research / Defense Science and Technology Group Grant – January 2017 to May 2019

*Best paper award at the 1st Annual Review and Workshop AFOSR – DSTG Co-Sponsored Research Program on Trusted Autonomy*

**“A Neurobehavioral Foundation for Affective Computing: Rat-Robot Brain-Computer Interfaces for Dynamic Interaction”** \$25,000

Kavli Institute for the Brain and Mind (KIBM) Innovative Research Grant – May 2016 to August 2017

**Grants Contributed To:**

**“Behavioral, Physiological, and Quantitative Models of Pro-Social Behavior”** \$1,500,000

NIMH R01 Research Grant – Aug 2016 to April 2019

**“Socially Situated Neuroscience: Creating a suite of tools for studying sociality and interoception”** \$300,000

NSF BRAIN EAGER Grant – Aug 2014 to April 2017

**Hofstra University** – Fall 2010 to Spring 2014

**Bachelor of Arts (BA); Triple Major in Psychology, History and Chinese Studies**

Summa Cum Laude; Phi Beta Kappa; Honors College Associate; Provost’s Scholar and Dean’s List, Vince Brown Scholarship

**GPA:** 3.87

## RESEARCH EXPERIENCE

**Principal Scientist, Autonomous Cardiac Control Systems (MaxWell Biomedical)** – Aug 2026 – Present

Developing a next generation autonomous control system for cardiac resynchronization therapy using techniques from dynamical systems theory, signal processing, electrophysiology, and machine learning.

**Machine Learning & 3D Computer Vision Consultant – UC San Diego (Contract)** Jan 2026 – Present

Consulting on the application of machine learning and 3D computer vision for behavioral analytics in educational environments.

Supporting research led by Professor Terry Jernigan and Professor Andrea Chiba. Advising on multi-camera calibration and 3D reconstruction pipelines for large-scale behavioral data. Training technical teams to deploy open-source pose estimation and computer vision tools

**Postdoctoral Fellow – Talmo Pereira and Eiman Azim Lab** – Aug 2022 – Present

Working with Talmo Pereira and Eiman Azim on deep 3D pose tracking in lab animals and massively parallel imitation learning in virtual biomechanical rodent models to understand the neural computations underlying motor control.

**Postdoctoral Fellow – Talmo Pereira and Tom Albright Lab** – Aug 2022 – Sept 2023

Working with Talmo Pereira and Tom Albright on deep pose tracking in lab animals and long term monitoring of human behavior in the Los Angeles County Museum of Art.

**Graduate Researcher – Andrea Chiba Lab** – Aug 2015 – Spring 2022

Working with Professor Andrea Chiba and Dr. Laleh Quinn on a rodent behavioral paradigm where a rodent learns how to control a robot in a goal oriented reinforcement learning paradigm.

**Visiting Scholar – Complex and Intelligent Systems Laboratory** – Mar 2016 – Nov 2017

Worked with Professor Janet Wiles at the University of Queensland on Recurrence Quantification Analysis / Topological Data Analysis.

**Graduate Researcher – Cognitive Neuroscience and Neuropsychology Lab** – Aug 2014 – Aug 2015

Worked with Professor Ayse Saygin on using electrophysiological techniques to investigate the neural basis of biological motion perception. I was responsible for designing experiments, analyzing data, programming android robots and experimental interfaces.

**Lab Rotation – Cognitive and Neural Dynamics Lab** – Mar 2015 – June 2015

Worked with Professor Bradley Voytek and a team of lab members to perform EEG data analysis using electrophysiological methods.

**Lab Assistant – Human and Artificial Learning Lab** – June 2013 – Aug 2014

Assisted Professor Oskar Pineño with designing Arduino physical computation applications. Constructing a low-cost EEG operated brain-computer interface and linking it with robotic devices.

**Editing Intern – “The History and Evolution of Psychology: A Philosophical and Biological Perspective”** – 2013– 2014

Paid academic internship with Professor Brian D. Cox, editing bibliographic information for a textbook published with Routledge.

**TEACHING/MENTORSHIP EXPERIENCE**

**Salk Institute for Biological Studies**

**Mentor – Salk Institute High School Summer Scholars Program** - Summer 2024

Acted as a mentor for high school research fellows supervising a 3D pose tracking project.

**Mentor – Salk Institute Summer Undergraduate Research Fellowship (SURF) Program** - Summer 2023, Summer 2024

Acted as a mentor for undergraduate research fellows supervising projects using deep learning for animal pose tracking.

**Teaching Assistant – SLEAP: Automating behavior quantification using deep learning** – February 2024

Assisted Talmo Pereira at COSYNE 2024 in Lisbon, Portugal in a tutorial workshop to get users started with quantifying animal behavior with the SLEAP deep learning framework.

**UC San Diego**

**Co-Instructor – DSC 180 – Deep Imitation Learning** – Fa 2022 - Wi 2023

Co-taught a Data Science capstone course with Dr. Talmo Pereira about developing deep reinforcement learning tools for rodent behavior.

**Instructor / Teaching Assistant – COGS 8 – Hands-On Computing** – Fa 2020, Wi 2015, 2021, Spr 2016, 2019, 2020, 2022

Taught Arduino programming in C and robotics from the perspective of cognitive science. Led as instructor Spr 2019 – Spr 2022

**Instructor / Teaching Assistant – COGS 100 – Cyborgs Now and In The Future** – Spr 2018, Fa 2018, Wi 2019

Assisted Professor Taylor Scott and Professor David Kirsh to teach an introduction to classical AI / cognitive science and human-computer interaction theory such as embodied, distributed and situated cognition in 2018. Lead as an instructor in Wi 2019.

**Instructor – Academic Connections – Introduction to Cognitive Science** – Summer 2015 - 2022

Worked with co-instructor Tom Donoghue to design and teach an introduction to cognitive science class for high school students, more than 40 hours of lectures spanning psychology, robotics, AI, neuroscience, linguistics, anthropology and philosophy for 7 years in a row.

**Instructor – Academic Connections – Hands-On Computing for Cognitive Science** – Summer 2020

Taught an online class for high school students about simulating nonlinear dynamical systems using Processing programming language.

**Co-Instructor – COMM 190 – Performing Cybernetics** – Spring 2017

**Teaching Assistant – COGS 17 – Neurobiology of Cognition** – Fall 2015 – **COGS 184 - Modeling the Evolution of**

**Cognition** – Winter 2015 – **COGS 179 – Cognitive Electrophysiology** – Fall 2019 – **COGS 170 – Brain Waves Across Scales** – Winter 2020

**REFERENCES**

**Publications and Conference Papers**

Leonardis, E. J., Oline, S. N., Tang, I., Park, J., Mimica, B., Maree, L., Afshar, S., Tye, K. M., Murthy, M., Falkner, A. L., Pereira, T. D. (In Prep). LUC3D: Browser-based 3D annotation and proofreading GUI for pose estimation and multi-animal tracking. To submit to *Nature Neuroscience*.

Bian, K., Leonardis, E. J., Yang, Y., Zhang, C., Wang, Y., & Pereira, T. D. (Accepted). Shaped by what's missing: Topological invariance simplification discovers behavioral transitions. *NeurIPS 2026: Symmetry and Geometry in Neural Representations Workshop*. Sydney, Australia.

Bian, K., Zhang, C., Yang, Y., Sirbu, A., Jha, A., Buchanan, K., Leonardis, E. J., Richards, B. A., Ölveczky, B. P., Linderman, S. W., Pereira, T. D. (Accepted). Using Symbols to Steer Actions in Embodied Biomechanical Agents. *NEmo: Neuro-Symbolic Embodied Intelligence Workshop*. Sydney, Australia.

Zhang, C., Yang, Y., Sirbu, A., Abe, E., Warnberg, E., Leonardis, E., Aldarondo, D. E., Lee, A., Prasad, A., Foat, J., Bian, K., Park, J., Bhatt, R., Saunders, H., Nagamori, A., Thanawalla, A. R., Huang, K. W., Plum, F., Beck, H., Flavell, S. W., Labonte, D., Richards, B. A., Brunton, B. W., Azim, E., Richards, B., Ölveczky, B. P., & Pereira, T. D. (Under Review). MIMIC-MJX: Neurobiomechanical emulation of animal behavior. *Nature: Methods*.

Leonardis, E., Nagamori, A., Yang, Y., Park, J., Saunders, H., Azim, E., Pereira, T. D. (2025) Massively Parallel Imitation Learning of Mouse Forelimb Musculoskeletal Reaching Dynamics. *NeurIPS 2025: Data on the Brain & Mind Concrete Applications of AI to Neuroscience and Cognitive Science Workshop*, San Diego, CA.

Jackson, I., Kohli, R., Leonardis, E., de Sa, V., Fei, S., Quinn, L., Lou, Y., & Chiba, A. (2025). Explore-Exploit Behaviors During Rat-Robot Interactions Optimize Social and Spatial Security. *IEEE International Conference on Development and Learning (ICDL)*. Prague, Czech Republic. September 16-19, 2025.

Maree, L., Afshar, S., Oline, S., Leonardis, E.J., Falkner, A. L., & T. D. Pereira (2024). Multi-view triangulation-enabled annotation for multi-animal 3D pose in SLEAP. *Measuring Behavior 2024: 13th International Conference on Methods and Techniques in Behavioral Research*, Aberdeen, Scotland.

Leonardis, E.J., Breston, L., Lucero-Moore, R., Sena, L., Kohli, R., Schuster, L., Barton-Gluzman, L., Quinn, L.K., Wiles, J., & Chiba, A.A. (2022). Interactive Neurorobotics: Behavioral and Neural Dynamics of Agent Interactions. *Frontiers in*

- Turner, M. (Chair), Eric Leonardis, Pelkey, J., Semenuks, A., Coulson, S., Adachi, I. & Forster, D. (2021). Conceptual Blending in Animal Cognition: A Comparative Approach. *43<sup>rd</sup> Annual Cognitive Science Society Conference 2021*, Vienna, AUT
- Breston, L., Leonardis, E. J., Quinn, L. K., Tolston, M., Wiles, J., & Chiba, A. A. (2021). Convergent Cross Sorting for Estimating Dynamic Coupling. *Scientific Reports*, 11(1), 1-10.
- Heath, S., Ramirez, C., Arnold, J., Olsson, O., Taufatofua, J., Pounds, P., Wiles, J., Leonardis, E., Gygi, E., Leija, E., Quinn, L., Chiba, A. (2018) PiRat: An autonomous framework for studying social behavior in rats and robots. *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2018)*, Madrid, Spain.
- Leonardis, E. (2017). Amygdala. & Hippocampus. *Encyclopedia of Animal Cognition and Behavior*. (Eds. Vonk, J. & Shackelford, T.K.) Springer.
- Leonardis, E. & Saygin, A. (2015). Humanoid Robots and the Social Brain: Ethical Implications. *Human-Robot Interaction (HRI) 2015: The Emergent Policy and Ethics of Human-Robot Interaction Workshop*, 10<sup>th</sup> ACM/IEEE International Conference.

### **Selected Poster Presentations**

- Leonardis, E. J., Nagamori, A., Siwanowicz, I., Sirbu, A., Thanawalla, A., Yang, Y., Park, J., Saunders, H., Huang, K. W., Zhang, C., Olveczky, B., Richards, B., Johnson, R., Azim, E., & Pereira, T. D. (2026). High-throughput imitation learning of reaching behavior with a musculoskeletal model of the mouse forelimb. Simulated Bodies: Whole Body Biomechanical Models, April 12–15, 2026, Howard Hughes Medical Institute – Janelia Research Campus, Ashburn, VA.
- Leonardis, E., Nagamori, A., Thanawalla, A., Yang, Y., Saunders, H., Gilmer, J., Zhang, C., Bian, K., Olveczky, B., Al Borno, M., Azim, E., Pereira, T. D. (2026). *Computational and Systems Neuroscience (COSYNE) 2024, Lisbon, Portugal*.
- Zhang, C., Sirbu, A., Yang, Y., Leonardis, E., Park, J., Prasad, A., Bian, K., Abe, E., Warnberg, E., Brunton, B., Richards, B., Olveczky, Pereira, T. D. (2026). MIMIC-MJX: Neuromechanical imitation of animal behavior enables flexible models of embodied control. *Computational and Systems Neuroscience (COSYNE) 2024, Lisbon, Portugal*.
- Leonardis, E.J., Yang, Y., Nagamori, A., Park, J., Bian, K., Barbano, A., Foat, J., Azim, E., Pereira, T.D. (2025) Behavior-Driven Musculoskeletal Modeling for Embodied Neural Control: Dextrous Forelimb Reaching in Mice. *Society for Neuroscience Conference 2025*, San Diego, CA.
- Yang, Y., Leonardis, E.J., Bian, K., Zhang, C., Azim, E., Öveczky, B., & Pereira, T. D. (2025). VNL-playground: fast and biologically realistic virtual environment for simulating animal behavior. *Society for Neuroscience Conference 2025*, San Diego, CA.
- Bian, K., Leonardis, E., Yang, Y., Zhang, C., Wang, Y., Öveczky, B., & Pereira, T. D. (2025). Topology-driven insights into naturalistic behavior from neuromechanical agent modeling. *Society for Neuroscience Conference 2025*, San Diego, CA.
- Jackson, I., Kohi, R., Lucero-Moore, R., Lou, Y., Quinn, L. K., Breston, L., Wiles, J., Chiba, A. A., Leonardis, E. (2025). Robotic exploratory control via subcortical oscillations. *Society for Neuroscience Conference 2025*, San Diego, CA.
- Leonardis, E.J., Barbano, A., Yang, Y., Nagamori, A., Foat, J., Gilmer, J., Al Borno, M., Azim, E., & Pereira, T.D. (2025). Examining the impact of biomechanical actuation on neural representations for embodied control and imitation learning for natural movement. *Computational and Systems Neuroscience (COSYNE) 2025, Montreal, Quebec, Canada*.
- Leonardis, E.J., Butler, D., Lee, A., Aldarondo, D.E., Öveczky, B., Azim, E., & Pereira, T.D. (2024). The impact of biomechanical actuation on neural embodied control. *Computational and Systems Neuroscience (COSYNE) 2024, Lisbon, Portugal*.
- Leonardis, E.J., Butler, D., Lee, A., Aldarondo, D.E., Öveczky, B., Azim, E., & Pereira, T.D. (2023). Examining the role of biomechanical actuation in neural embodied control. *Simulated Bodies: Whole Body Biomechanical Models*. Howard Hughes Medical Institute - Janelia Research Campus, Ashburn, VA.
- Maree, L., Leonardis, E., Gepstein, S., Pfaff, S., Metallo, C., & Pereira, T. (2023). Quantifying behavior using deep learning. *Society of Biological Psychiatry Annual Meeting*, San Diego, CA. *Biological Psychiatry*, 93(9), Supplement 7.
- Leonardis, E. J., Semenuks, A., Sims, G., & Coulson, S. (2022). How the structures of umwelts and signs are related. In *Proceedings of the Joint Conference on Language Evolution (JCoLE 2022)*.
- Leonardis E.J., Breston L., Lucero-Moore R., Kohli R., Barton-Gluzman, M., Aguilar-Rivera, L., Quinn L.K., Wiles J., & Chiba A.A. (2022). Brain and Body Coupling in Neural Circuitry Underlying Social Assessment. *Society for Neuroscience Conference 2022*, San Diego, CA.
- Mullane, M. D., Cooper, H., Lindner, T., Leonardis, E. J., & Chiba, A. A. (2018). Perceiving emotional sounds (MAARI): Individual differences, prior learning and context. *Society for Neuroscience and Society for Social Neuroscience Conference 2018*, San Diego, CA.
- Heath, S., Ramirez, C., Arnold, J., Olsson, O., Taufatofua, J., Pounds, P., Wiles, J., Leonardis, E., Gygi, E., Leija, E., Quinn, L., Chiba, A. (2018) PiRat: An autonomous rat-sized robot as a social companion for studying social behavior in rats using real-time tracking. *Society for Neuroscience Conference 2018*, San Diego, CA.
- Leonardis, E., Heath, S., Wiles, J., Chiba, A. A., Quinn, L. K. (2016). Social Investigation of Conspecifics and Robots: Oscillatory Neural Dynamics. *Society for Neuroscience Conference 2016*, San Diego, CA.
- Leonardis, E., Heath, S., Wiles, J., Quinn, L. K., Chiba, A. A. (2016). A Social Brain-Computer Interface for Rat-Robot Interactions. *Network for The Science of Learning Meeting 2016*, Arlington, VA.

### **Conference Talks + Invited Lectures**

- Leonardis, E. (2023, 2024, 2026). Computational Ethology and Embodied AI. Invited Guest Lecture for *COGS 13 Field Methods: Cognition in the Wild* with Professor Federico Rossano. UCSD, La Jolla, CA.
- Leonardis, E. (2025). Science for Screenwriting and Film. Invited Guest Lecture for STS 497 Advanced Special Topics in Science, Technology and Society Topic: Science Communication with Professor Yelena Gluzman. University of Alberta. Edmonton, Canada.
- Leonardis, E. (2025) The Elephant Man and Disability. *American Academy of Neurology Meeting and Dr. Ali Christy's History of Neurology and Film Class*, April 8th. Digital Gym Cinema, San Diego, CA.
- Leonardis, E. (2024) Body Horror and the Brain. *Neurohumanities Network Event*. Harvard Medical School. Boston, MA. Virtual.
- Leonardis, E. (2024) Explaining AI. Invited Public Lecture at *Hofstra University Cultural Center*, April 16th, Hofstra University, Hempstead, NY.
- Leonardis, E. (2024). AI Ethics. Invited Guest Lecture for *CGS 010 Introduction to Cognitive Science* with Professor John Teehan, April 16th, Hofstra University, Hempstead, NY.
- Leonardis, E. (2024). Deep Learning for 2D and 3D Pose Estimation. Invited Guest Lecture for *CSC 158 Artificial Intelligence* with Professor Simona Doboli, April 15th, Hofstra University, Hempstead, NY.
- Leonardis, E., Semenuks, A., & Coulson, S. (2021). What is indexical and iconic in animal blending? In *Conceptual Blending in Animal Cognition: A Comparative Approach*. 43<sup>rd</sup> Annual Cognitive Science Society Conference 2021, Vienna, AUT
- Leonardis, E. (2020). The Misuse of Algorithms in Data Science. Invited Guest Lecture for *COMM 164 Behind the Internet: Invisible Geographies of Power and Inequality* with Instructor Yelena Gluzman, UCSD, La Jolla, CA.
- Leonardis, E. (2018, 2019). Cyborg as Post-Human. Invited Guest Lecture for *COGS 100 Cyborgs Now and In The Future* with Instructor Michael Allen, UCSD, La Jolla, CA.
- Leonardis, E. (2017). Rat-Robot Brain Computer Interfaces for Dynamic Interactions. *Kavli Institute for the Brain and Mind Annual Symposium on Innovative Research*. La Jolla, CA.
- Leonardis, E. (2017). Rats, Robots, Respiration and Rhythm. *Cognition at Work: UCSD Cognitive Science Student Association's 11th Annual National Cognitive Science Conference*, UCSD, April 9<sup>th</sup>, La Jolla, CA.
- Leonardis, E. (2017). Historical Examples of Misapplications in Data Science. Invited Guest Lecture on Data Ethics for COGS 9 Introduction to Data Science with Dr. Jason Fleischer.

### **Technical Demonstrations**

- Leonardis, E., D'Amico, A., Guerin, S., Verhoef, T., & de Sa, V. (2018). PenguinBird OpenBCI DIY Educational Platform. *The Equity Journey: Investing in the Whole Learner. Grantmakers For Education Conference 2018*. Coronado, CA
- Leonardis, E., Mousavi, M., Miller-Rigoli, C., Cooper, H., Contreras, F., & Verhoef, T. (2018). PenguinBird: Dancing Robot Brain-Computer Interface. *IBM Artificial Intelligence for Healthy Living Center (AIHL) SmartHome Demonstration* at Calit2, UCSD, La Jolla, CA.
- Leonardis, E., Heath, S., Wiles, J., Quinn, L. K., Chiba, A. A. (2016). Brain-Computer Interfaces (BCI) for Social Interaction and Animal Models. *Temporal Dynamics of Learning Center Demo Session*, January 28<sup>th</sup> in La Jolla, CA.

### **Science Communication, Radio/Podcasts, and Exhibited Art**

- Leonardis, E. (Panelist), Mendez Gandica, B., Jones, L., Brown, A., Riek, L., & Decker, A. (Moderator). (2025). The Evolution of A.I. through Film and in Reality. *San Diego Comic-Con International*. San Diego, CA.
- Leonardis, E. (2025). Animal Communication in Babe: Pig In The City. *Science on Screen*. Digital Gym Cinema, San Diego, CA.
- Leonardis, E. (2024). Neuroscience of the New Flesh: Body Horror in Your Brain. *Nerd Nite SD*. Ken Club, San Diego, CA.
- Leonardis, E. (2023). Neuroscience of Fear and Zombie Languages. *Secret Morgue 4: Zombie Autopsy*. Comic Con Museum. San Diego, CA.
- Farokhmanesh, M. (2023). What Creepy Video Game Sounds Do to Your Brain. *WIRED*. February 7th.
- Leonardis, E. (2019). Cats and Baboons. *Secret Morgue 3.1: When Animals Attack*. Comic Con Museum. San Diego, CA.
- Accomando, B. (2020). Global Pandemic Film Primer with Eric Leonardis. *Cinema Junkie Podcast on KPBS*. San Diego, CA.
- Accomando, B. (2020). Pop Culture, Neuroscience, And COVID-19 with Eric Leonardis. *KPBS Midday Edition*. San Diego, CA.
- Leonardis, E. (2019). Brain-Computer Interfaces: Applications and Ethical Conundrums. *National Geographic Brain Games and San Diego Nerd Nite at San Diego Comic Con*. Hotel Solamar, San Diego, CA.
- Leonardis, E. (2019). Introducing David Cronenberg's The Fly. Reel Science 3.0 at Digital Gym Cinema with SD Film Geeks and San Diego Natural History Museum. San Diego, CA. Radio Promo on *KPBS Midday Edition* and *Cinema Junkie Podcast*.
- Gluzman, Y. & Leonardis, E. (2015). Their Position. *The Ephemeral Objects Exhibit*. San Diego Art Institute. San Diego, CA.

### **Reviewer**

*NeurIPS: Data on the Brain and Mind Workshop*  
*Scientific Data (Nature)*  
*Communications Physics (Nature)*  
*Journal of Experimental Psychology: Applied*  
*Computational and Systems Neuroscience (COSYNE) Conference*  
*Frontiers in Cellular Neuroscience, Section on Cellular Neurophysiology*  
*Frontiers in Bioengineering and Biotechnology, Section on Bionics and Biomimetics: Hypothesis and Theory*

**Scientific Consulting for Film**

*National Academy of Sciences Science and Entertainment Exchange  
NYU Tisch Film and Television  
Sloan Foundation*

**Community**

*Underrepresented Minorities Advancing Scientific Engagement (URMASE) Events Chair  
Salk (EDGE) Elevate Diversity in Graduate Education Program*